Before you commit a large experiment, find out whether it can teach you enough.

EvoForecast is a reproducible digital wind tunnel. It asks a narrower and more useful question than whether evolution is predictable. Can a proposed observation programme tell prospective forecasts apart before a costly physical programme is built?

5,600/5,600

latest blind synthetic SLiM 5.2 runs completed, none failed

0/60

latest forward-looking cells that reached the indicative 50 percent bar

0/120

latest experiment plans classified robustly best

Claim boundary. These are synthetic software and experimental-design results. They are not empirical Daphnia magna forecast skill, GT01 or G1 qualification, or biological validation, and they carry no endorsement, partnership, or funding claim.

The choice that sets up every other choice.

Species, generations, populations, starting diversity, conditions, assays, and automation form one coupled design. The wind tunnel tests which combinations could resolve a forecast and which only add observations without telling designs apart.

Claims EF-001 to EF-006 · exact source hashes in audit/source_manifest.json

An auditable chain from parameters to a scored forecast.

REGISTERED PARAMETERS	SLiM 5.2 WORLD	LATENT STATE
OBSERVATION MODEL	MODEL FORECAST	IMMUTABLE DEPOSIT
INDEPENDENT REVEAL	PROPER SCORE	DESIGN / ABSTAIN

What the latest challenge says

The genome-plus-phenome track improved its mean CRPS by 47.5 percent over the simplest baseline, aggregated across 35 registered contexts. Its ninety percent prediction range still held the truth only eight times in ten. No forward-looking cell reached the combined bar. Average accuracy, calibration of that range, population-level identifiability, and transfer are separate tests.

> **Interpretation.** This is informative synthetic design evidence. It is not biological forecast skill.

Why controls matter

A corrected favorable marker control produced a 17.0 percent genome-versus-pedigree gain. A matched adverse control produced minus 9.2 percent. The opposite signs show the workflow can expose failure rather than manufacture a genomic advantage.

> **Supported claim.** The software control behaves differently under registered favorable and adverse conditions.

The signature instruments

The build-the-experiment view returns registered gain, coverage, uncertainty, and combined-bar probability for a chosen track, population count, and horizon. The measurement-plan frontier places forecast value against modeled resource use for named packages. Both are precomputed and hash-bound. Browser controls only select rows.

Claims EF-003 to EF-010 and EF-013 · no browser fitting, simulation, or post-outcome optimization

The software claim is solid. The biology claim is not, yet.

1 · Software proof	SUPPORTED · SYNTHETIC
2 · Indicative design evidence	CONDITIONAL
3 · Empirical qualification	BLOCKED
4 · Operational forecasting	FUTURE

What the theoretical cutaway proposes

A post-RC01 GT01 design would connect diploid ten-chromosome standing founder variation and sexual recombination to phenotype, ecology, sampling, assays, blind forecasts, and proper scoring. It remains proposed and uncalibrated. The frozen RC01 production model remains clonal.

What remains blocked

Exact founder and genome identity, accepted G0 and G1 evidence, real assay performance, material custody, independent prospective reveal, empirical uncertainty analysis, safety authority, and external reproduction.

Run one small round together.

Bring one candidate keystone organism, one budget you can actually spend, and one held-out challenge. We rehearse it in the wind tunnel, then run the smallest real round that can estimate calibration and forecast accuracy under an independent reveal. Only after that first round clears every gate do we stage up toward a network of interacting species around the keystone organism.

EvoForecast does not claim that evolution is predictable. It shows what a real experiment would have to measure to find out, and it reports plainly when a design is not ready.

Claims EF-011 to EF-017 · evidence bundle, claim registry, and lineage DAG accompany this note.